

Language contact is not consistently associated with increased feature economy

Muhammad Rehan David Goldstein

rehanmuh@g.ucla.edu dgoldstein@humnet.ucla.edu

Introduction Feature economy, the maximization of segmental contrasts per contrastive feature, has long been considered an organizing principle of consonant inventories (Jakobson, 1941; Ohala, 1980; Clements, 2003a,b), but it is increasingly viewed as an emergent property of sound systems (Martinet, 1955; Marsico et al., 2004; Blevins, 2005; Schenck and Pfiffner, 2026; Narkar, 2026). Segmental borrowing has been proposed as one source of this emergence because borrowing can fill structural gaps in recipient inventories (Weinreich, 1953; Maddieson, 1985; Eisen, 2019; Martin et al., 2022). Under the gap-filling hypothesis, regions with extensive borrowing should exhibit elevated feature economy. We test this prediction by asking whether, after accounting for shared ancestry, geographic patterns in feature economy are consistent with this expectation.

Data and methods We fit hierarchical Bayesian models to PHOIBLE consonant inventories (Moran and McCloy, 2019) on the global phylogeny of Bouckaert et al. (2026) and the polygons of Ranacher et al. (2026). The response variable is the excess number of active feature dimensions above the theoretical minimum required for a consonant inventory of that size. All models include the log-transformed consonant inventory size of a language as a fixed effect and the original inventory source (e.g., AA, LAPSyD) as a varying effect. The spatiophylogenetic models incorporate a phylogenetic varying effect and differ only in their spatial component: a Besag model over language adjacency, a Gaussian process over geographic distance, or a combination of both.

Results Across all models, inventories require more active feature dimensions than the theoretical minimum for inventories of their size. However, larger inventories lie closer to this minimum than smaller inventories (Fig. 1). The model with phylogenetic and both adjacency- and distance-based spatial effects has the highest predictive accuracy. The low correlation between the two spatial components ($r \approx 0.23$) indicates that adjacency and geographic distance capture complementary rather than redundant spatial structure. The overall spatial effect is modest, however (Fig. 2), and inconsistent with the prediction of the gap-filling hypothesis. Although several regions widely recognized as contact zones (Australia, northern Eurasia, and parts of Amazonia) are more economical than predicted, others (South Asia, mainland Southeast Asia, and much of sub-Saharan Africa) exhibit lower feature economy than predicted.

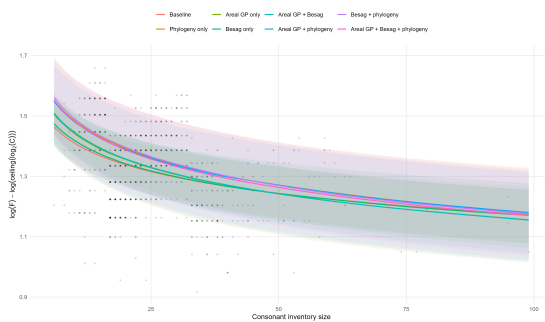


Figure 1: Fitted economy curves under each model

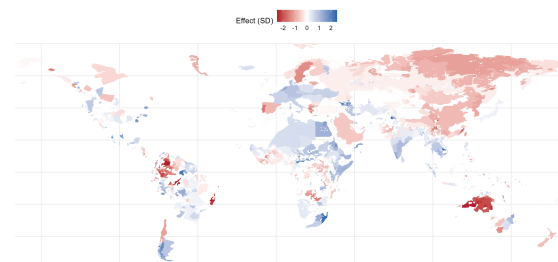


Figure 2: Combined spatial effect. Red indicates greater economy than predicted, blue less.

Discussion Previous work has emphasized the potential for borrowing to increase feature economy by filling structural gaps. By contrast, our analyses suggest that gap-filling is insufficient to explain global geographic patterns in feature economy. One possible explanation is that borrowing outcomes are more heterogeneous than such an account predicts. Borrowing may introduce segments that are rare across languages (Hagège and Haudricourt, 1978; Cline, 1984) or unusual for the borrowing language relative to its segmental inventory (Eisen, 2019; Moran et al., 2024). Alternatively, borrowed segments may be mapped onto existing phonemes rather than phonologized as new contrasts (Paradis and LaCharité, 1997; Kang, 2011). These outcomes provide a plausible explanation for why contact zones exhibit both lower and higher feature economy. More generally, our findings suggest that the effects of language contact on consonant inventories may be mediated by segmental markedness and perceptual assimilability.

References

- Juliette Blevins. Some problems with feature economy. Labial fricatives, lone fricatives, and lost labials. In *Seoul Linguistics Forum*, pages 211–230. Language Education Institute, Seoul National University, 2005.
- Remco Bouckaert, David Redding, Artur Trebski, Oliver Sheehan, Russell Gray, Kate E. Jones, and Quentin Atkinson. Global language diversification is linked to socio-ecology and threat status. SocArXiv preprint, March 2026. URL https://osf.io/preprints/socarxiv/f8tr6_v2. Preprint.
- George Nick Clements. Feature economy in sound systems. 20(3):287–333, 2003a. doi: 10.1017/S095267570400003X.
- George Nick Clements. Feature economy as a phonological universal. In María Josep Solé, Daniel Recasens, and Joaquín Romero, editors, *Proceedings of the 15th International Congress of Phonetic Sciences (ICPhS 15)*, pages 371–374. Futurgraphic, 2003b. URL <https://shs.hal.science/halshs-00182695>.
- D. Cline. What does a sound have to do to get borrowed into a language? Manuscript, Department of Linguistics, University of California, Los Angeles, 1984.
- Elad Eisen. The typology of phonological segment borrowing. Master’s thesis, The Hebrew University of Jerusalem, 2019.
- Claude Hagège and André-Georges Haudricourt. *La phonologie panchronique*. Presses Universitaires de France, Paris, 1978.
- Roman Jakobson. *Kindersprache, Aphasie und allgemeine Lautgesetze*. Almqvist & Wiksell, Uppsala, 1941.
- Yoonjung Kang. Loanword phonology. In Marc van Oostendorp, Colin J. Ewen, Elizabeth Hume, and Keren Rice, editors, *The Blackwell Companion to Phonology*, volume IV. Wiley-Blackwell, Oxford, 2011. doi: 10.1002/9781444335262.wbctp0095.
- Ian Maddieson. Borrowed sounds. *UCLA Working Papers in Phonetics*, 61:51–64, 1985. URL <https://escholarship.org/uc/item/6z50c4zv>.
- Egidio Marsico, Ian Maddieson, Christophe Coupé, and François Pellegrino. Investigating the “hidden” structure of phonological systems. *Proceedings of the Annual Meeting of the Berkeley Linguistics Society*, 30(1):256–267, 2004. doi: 10.3765/bls.v30i1.3427.
- Alexander Martin, Marieke van Heugten, René Kager, and Sharon Peperkamp. Marginal contrast in loanword phonology: Production and perception. *Laboratory Phonology*, 13(1), 2022. doi: 10.16995/labphon.6454.
- André Martinet. *Économie des changements phonétiques. Traité de phonologie diachronique*. Francke, 1955.
- Steven Moran and Daniel McCloy, editors. *PHOIBLE 2.0*. Max Planck Institute for the Science of Human History, Jena, 2019. URL <https://phoible.org/>.
- Steven Moran, Elad Eisen, Dmitry Nikolaev, and Eitan Grossman. Operationalizing borrowability: Phonological segments as a case study. *Language*, 100(4):671–698, 2024. doi: 10.1353/lan.2024.a947038.
- Jahnvi Narkar. Optimal stop inventories. Laryngeal settings, 2026. PhD Dissertation.
- John J. Ohala. Chairman’s introduction to symposium on phonetic universals in phonological systems and their explanation. In *Proceedings of the Ninth International Congress of Phonetic Sciences*, pages 184–185, Copenhagen, 1980. Institute of Phonetics, University of Copenhagen.

- Carole Paradis and Darlene LaCharité. Preservation and minimality in loanword adaptation. *Journal of Linguistics*, 33(2):379–430, 1997. doi: 10.1017/S0022226797006786.
- Peter Ranacher, Robert Forkel, Nour Efrat-Kowalsky, Matthias Urban, Antonia Hehli, Micha Franz, Gregory Biland, Aaron Kreienbühl, Alba Hermida Rodríguez, Matheus C. B. C. Azevedo, James Giebler, Takuya Takahashi, Nico Neureiter, Rik van Gijn, Meeli Roose, Outi Vesakoski, Robert Weibel, Gereon Kaiping, and Sietze Norder. Glottography: an open-source geolinguistic data platform for mapping the world’s languages. *Journal of Open Humanities Data*, 2026. doi: 10.5334/johd.459.
- Kai Schenck and Alexandra Marie Pfiffner. PHOIBLE inventories suggest that diachrony contributes to the appearance of feature economy. *Proceedings of the Linguistic Society of America*, 11(1):6124, 2026. doi: 10.3765/plsa.v11i1.6124.
- Uriel Weinreich. *Languages in Contact: Findings and Problems*. Linguistic Circle of New York, New York, 1953. Publications of the Linguistic Circle of New York 1.